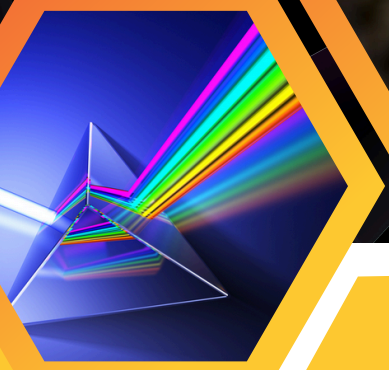






SSLC 2025

PHYSICS

learning material.....

Sanjeevkumar Isaraddi

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CHAPTER-9

LIGHT : REFLECTION AND REFRACTION

TOPIC-1 REFLECTION OF LIGHT

When light falls on a body, it may be absorbed, may be transmitted or light may get reflected back to the same medium.

The phenomenon of bouncing back off the light rays in the same medium is called **reflection.**

Laws of Reflection:

(i) The incident ray, the reflected ray and the normal, all lie on the same plane at the point of incidence.

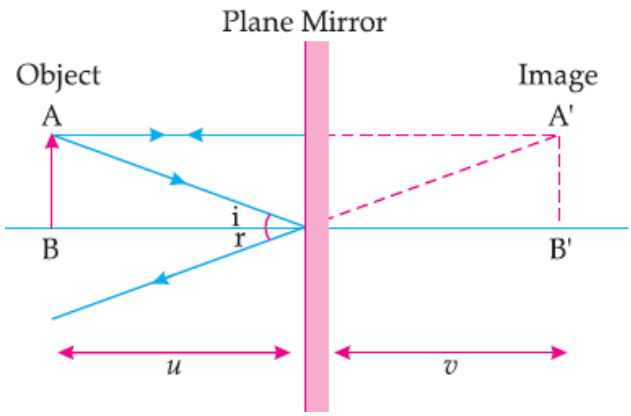
(ii) The angle of incidence is equal to the angle of reflection.

Nature of the images formed by the mirrors :

The image which can be obtained on the screen is called **real image** and which can not be obtained on the screen is called **virtual image**.

The real images are always inverted and virtual images are erected.

Characteristics of image formed by plane mirror:



- Image formed by a plane mirror is always virtual and erect.
- The size of the image is equal to that of the object.
- The image formed is as far behind the mirror as the object is in front of it.
- The image is laterally inverted.

SPHERICAL MIRRORS

The reflecting surface of a spherical mirror may be curved inwards or outwards.

A spherical mirror, whose reflecting surface is curved inwards, that is, faces towards the centre of the sphere, is called a **concave mirror.**

A spherical mirror whose reflecting surface is curved outwards, is called a **convex mirror.**

Key terms :

Pole : The centre of the reflecting surface of a spherical mirror is called the pole. The pole is usually represented by the letter P.

Centre of curvature : The reflecting surface of a spherical mirror forms a part of a sphere. The center of the sphere is called the centre of curvature. It is represented by the letter C.

The centre of curvature of a concave mirror lies in front of it. However, it lies behind the mirror in case of a convex mirror.

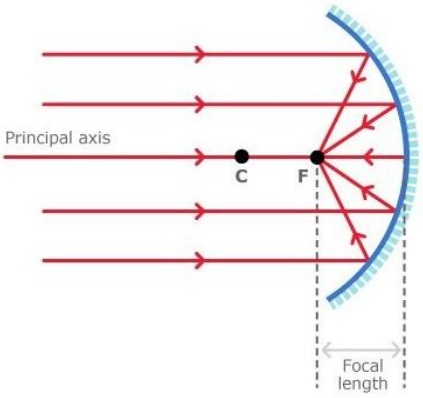
Radius of curvature : The radius of the sphere of which the reflecting surface of a spherical mirror is called the radius of curvature of the mirror. It is represented by the letter R.

The distance between P and C is the radius of curvature.

Principal axis : Imaginary straight line passing through the pole and the centre of curvature of a spherical mirror is called the principal axis.

Principal focus of the concave mirror :

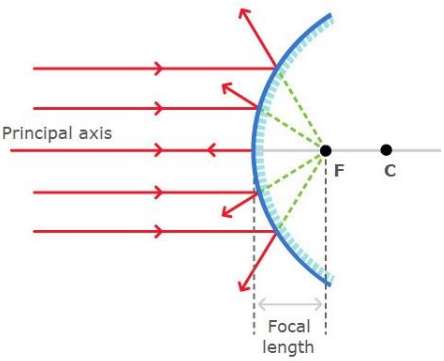
Rays parallel to the principal axis after falling on a concave mirror they are all converge and meet at a point on the principal axis of the mirror. This point is called the principal focus of the concave mirror. It is represented by the letter F.



The distance between the pole and the principal focus of a spherical mirror is called the **focal length**. It is represented by the letter f.

Principal focus of the convex mirror :

The principal focus of convex mirrors lies behind the mirror therefore light appears to be diverging when it gets reflected from the convex mirror.



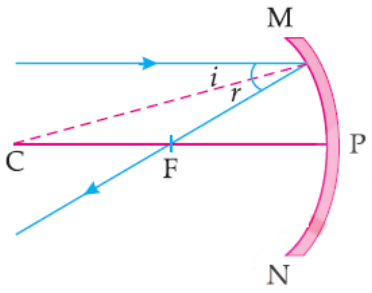
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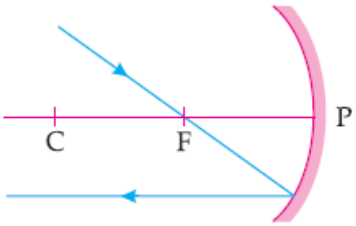
Aperture : The diameter of the reflecting surface of spherical mirror is called its aperture.

The radius of curvature is found to be equal to twice the focal length. ie, R = 2f .

Rules for making ray diagrams by concave mirror:

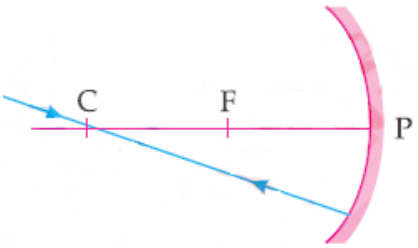
(i) A ray parallel to the principal axis will pass through the principal focus, after reflection.

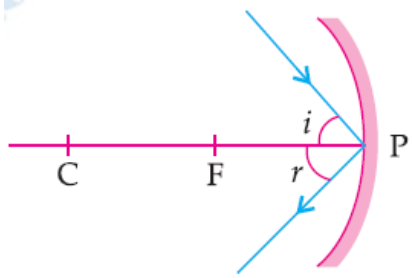




(ii) A ray passing through the principal focus of concave mirror will be parallel to principal axis after reflection.

(iii) A ray of light passing through the centre of curvature of a concave mirror is reflected back along the same path as it is a normally incident ray at the mirror surface.





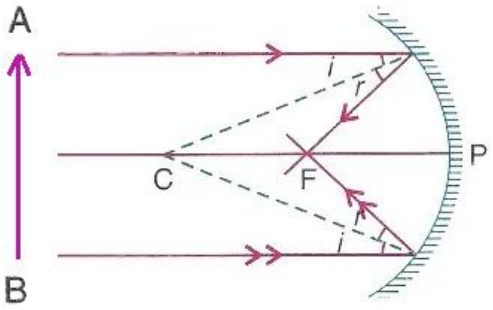
(iv) A ray incident obliquely to the principal axis of a concave mirror is reflected obliquely making equal angle with the principal axis.

Image formation by a concave mirror for different positions of the object:

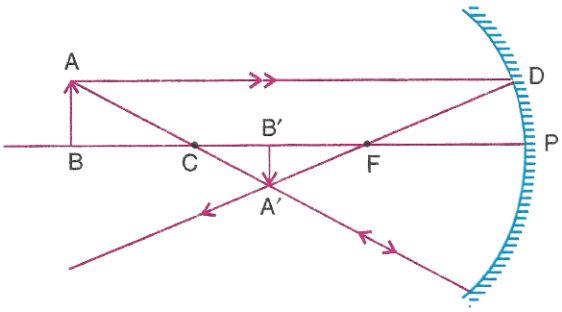
Position of Object	Position of Image	Size of Image	Nature of Image
At infinity	At the focus F	Highly diminished/ point-sized	Real and inverted
Beyond C	Between F and C	Smaller in size	Real and inverted
At C	At C	Same size	Real and inverted
Between C and F	Beyond C	Enlarged	Real and inverted
At F	At infinity	Highly enlarged	Real and inverted
Between P and F	Behind the mirror	Enlarged	Virtual and erect

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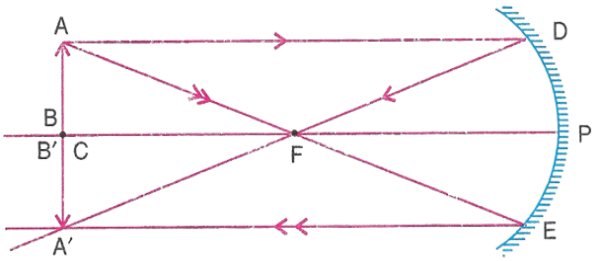
Object placed at Infinty



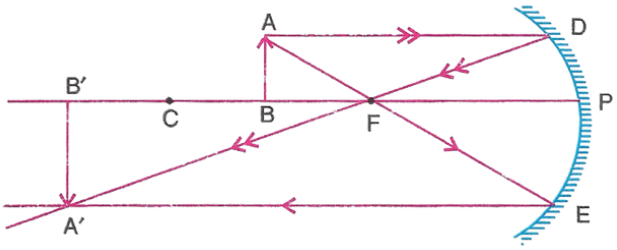
Object placed beyond C



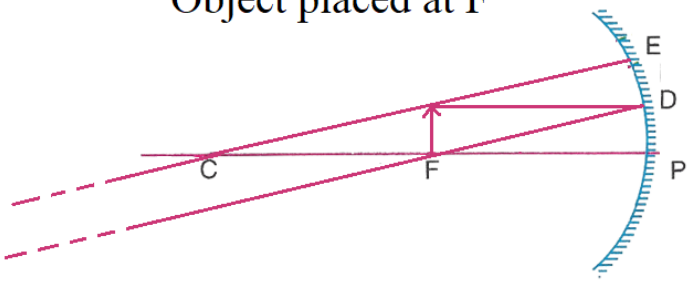
Object placed at C



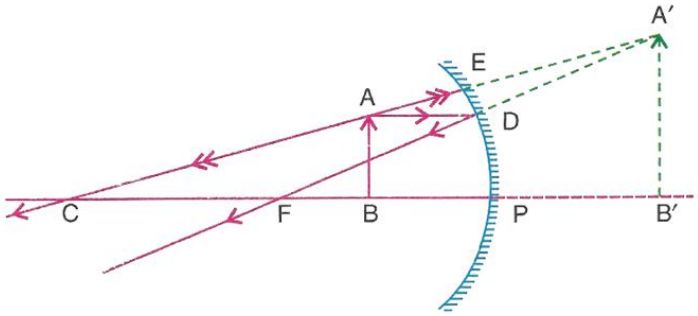
Object placed between C and F



Object placed at F



Oject placed between F and P



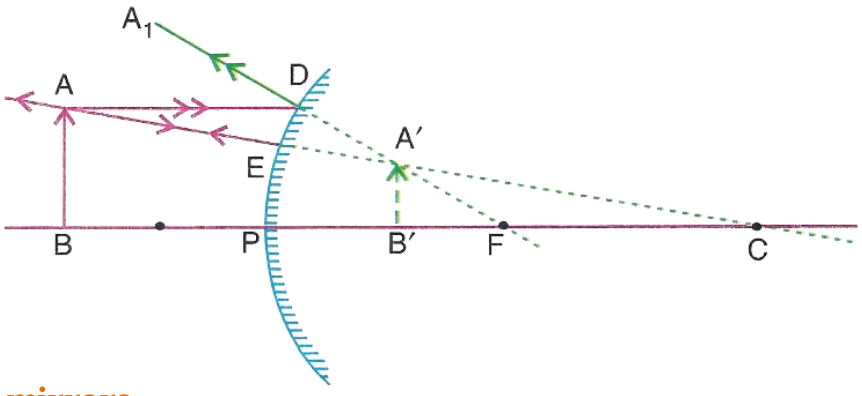
Uses of concave mirrors

i. Concave mirrors are commonly used in torches, search-lights and vehicles headlights to get powerful parallel beams of light.
ii. They are often used as shaving mirrors to see a larger image of the face
iii. The dentists use concave mirrors to see large images of the teeth of patients.
iv. Large concave mirrors are used to concentrate sunlight to produce heat in solar furnaces.

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Image formation by a convex mirror for different positions of the object:

Position of Object	Position of Image	Size of Image	Nature of Image
At infinity	At the focus F, behind the mirror	Highly diminished/ point-sized	Virtual and erect
Between infinity and the pole P of the mirror	Between P and F, behind the mirror	Diminished	Virtual and erect



Uses of convex mirrors

- Convex mirrors are commonly used as rear-view (wing) mirrors in vehicles,because they always give an erect, though diminished, image.
- Also, they have a wider field of view as they are curved outwards. Thus, convex mirrors enable to view much larger area.

Mirror formula and Sign conventions

Mirror Formula:

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

Where, v = Image distance
u = Object distance
f = Focal length

Magnification of Spherical Mirrors:

It is the ratio of the height of image to the height of object.

$$m=\frac{Height\ of\ Image}{Height\ of\ Object}=\frac{h'}{h}$$

$$m=\frac{h'}{h}=-\frac{v}{u}$$

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Key points :

- If 'm' is negative, image is real.
- If 'm' is positive, image is virtual.
- If m = 1, i.e., image is equal to object.
- If $h^i > h$ then $m > 1$ i.e., image is larger than object.
- If $h^i < h$ then $m < 1$ i.e., image is smaller than object.
- Magnification of plane mirror is always + 1.

  '+' sign indicates virtual image.

  '1' indicates that image is equal to object's size.
- If 'm' is '+ve' and less than 1, it is a convex mirror.

Exercise :

An object is kept on the principal axis of a concave mirror of focal length 12 cm. If the object is at a distance of 18 cm from the mirror, calculate the image distance. Determine the nature of the image formed by calculating the magnification produced by the mirror.

Solution :

Focal length f = -12cm , object distance u = -18cm , image distance v = ? ,

Lens formula : $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ OR $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$$\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18}$$

$$\frac{1}{v} = \frac{1}{-12} + \frac{1}{18} = \frac{-3+2}{36}$$

$$\frac{1}{v} = -\frac{1}{36}$$

$$v = -36\ cm$$

Magnification :

$$m = \frac{-v}{u} = \frac{-(-36)}{-18}$$

m = -2

Nature of the image : Real , inverted and larger than object

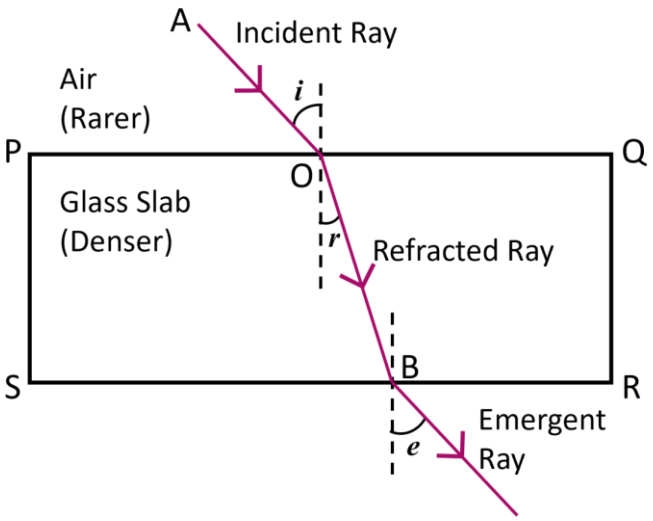
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TOPIC-2 REFRACTION OF LIGHT

The phenomenon of change in the path of light from one medium to another is called **refraction** of light.

The cause of refraction is the change in the speed of light as it goes from one medium to another medium.

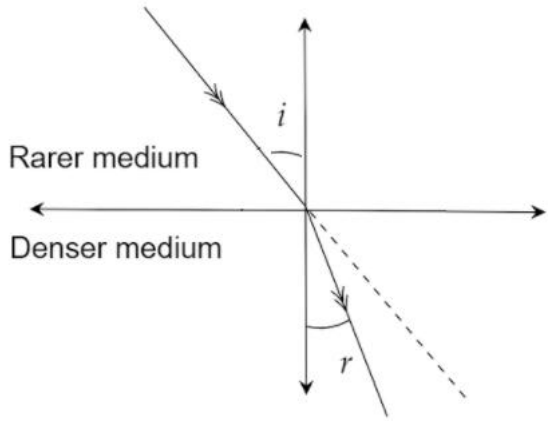
Refraction through a Rectangular Glass Slab

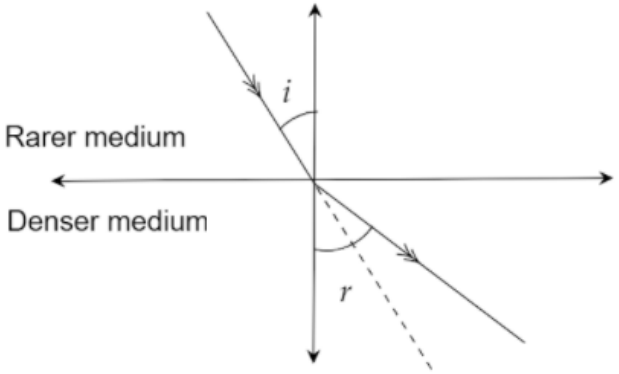


The angle formed between the incident ray and the normal is called angle of incidence and the angle formed between the refracted ray and the normal is called angle of refraction.

When a ray of light passes from a rarer medium to a denser medium, it bends towards the normal. Also, the angle of incidence is greater than the angle of refraction.

When a ray of light passes from a denser medium to a rarer medium, it bends away from the normal. Also, the angle of incidence is less than the angle of refraction.





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Laws of refraction:

First law: The incident ray, the refracted ray and the normal at the point of incidence all lie on the same plane.

Second law: The ratio of the sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given color and for the given pair of media. This law is also known as Snell's law of refraction.

$$\frac{Sin\ i}{Sin\ r} = n$$

Refractive index (n):

The ratio of speed of light in a given pair of media, is called refractive index

$$n = \frac{Velocity\ of\ light\ in\ medium\ 1}{Velocity\ of\ light\ in\ medium\ 2}$$

Absolute Refractive Index:

Refractive index of a medium with respect to vacuum or air.

$$n = \frac{C}{v} \quad \text{where } C = 3x10^8 ms^{-1}$$

- Refractive index of diamond is the highest till date. It is 2.42. It means speed of light is 2.42 times less in diamond than in vacuum.
- As the value of refractive index increases, the speed of light decreases.

Refractive index of glass is 1.50. Then speed of light in the glass would be -

Refractive index of glass $n_m = 1.50$

Speed of light in vacuum $c = 3 \times 108\ ms^{-1}$.

Speed of light in glass $v = \frac{c}{n_m} = \frac{3 \times 10^8}{1.50}$

$= 2 \times 10^8\ ms^{-1}$

Refraction by Spherical Lenses

A transparent material bound by two surfaces, of which one or both surfaces are spherical, forms a lens.

Convex lens :

- A lens which have two spherical surfaces, bulging outwards is called convex lens.
- It is thicker at the middle as compared to the edges.
- Convex lens converges light rays hence, convex lenses are also called converging lenses.

Concave lens :

- A lens which have two spherical surfaces and curved inwards is called concave lens.
- It is thicker at the edges than at the middle.
- Concave lenses diverge light rays. Such lenses are also called diverging lenses.

The surfaces of the lens forms a part of a sphere. The centers of these spheres are called **centers of curvature** of the lens. The centre of curvature of a lens is usually represented by the letter C. Since there are two centers of curvature, we may represent them as C_1 and C_2.

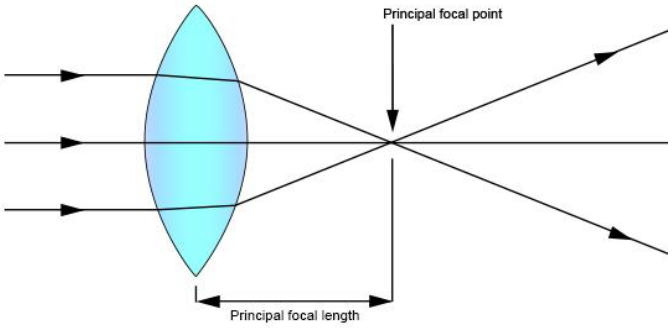
An imaginary straight line passing through the two centers of curvature of a lens is called its **principal axis.**

The central point of a lens is its **optical center**. It is represented by the letter O. A ray of light through the optical centre of a lens passes without suffering any deviation.

The effective diameter of the circular outline of a spherical lens is called its **aperture.**

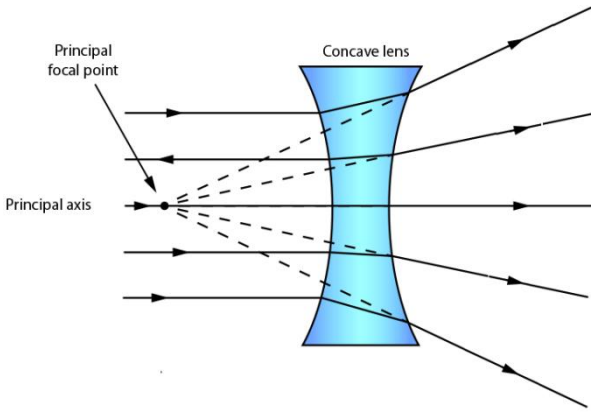
Principal focus of a convex lens :

Rays of light parallel to the principal axis after refraction are converging at a point on the principal axis. This point on the principal axis is called the principal focus of the lens.



Principal focus of a concave lens :

Rays of light parallel to the principal axis after refraction are appearing to diverge from a point at the principal axis is called the principal focus of the concave lens.



Letter F is usually used to represent principal focus. However, a lens has two principal foci. They are represented by F_1 and F_2.

The distance of the principal focus from the optical centre of a lens is called its **focal length.** The letter f is used to represent the focal length.

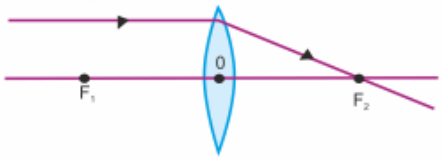
Image Formation by Convex Lens

Rules for the Formation of Image by the Lenses.

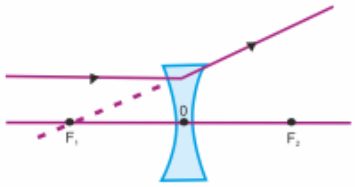
1) In both the lenses a ray of light passing through the optical center goes without any deviation.



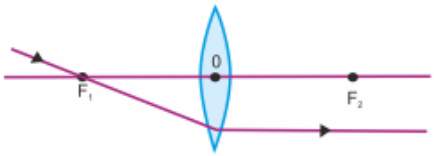
A ray of light parallel to the principal axis after refraction passes through the focus on the other side of the lens.



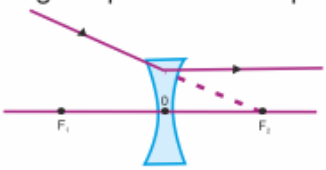
A ray of light appears to diverge from the focus on the same side of the lens.



A ray of light passing through the focus after refraction goes parallel to the principal axis.



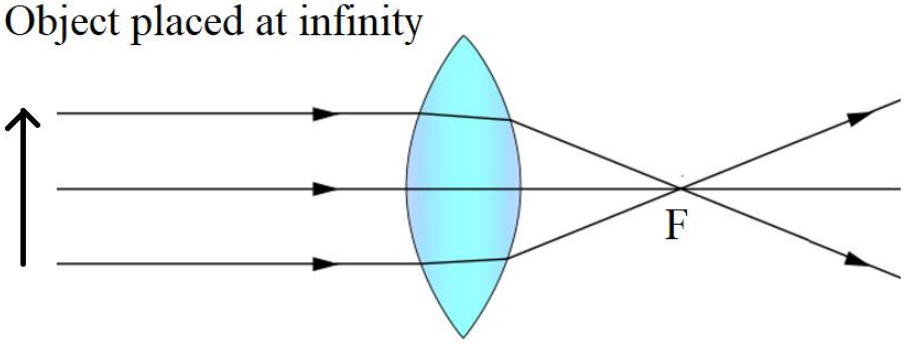
A ray of light directed towards the focus after refraction goes parallel to the principal axis.



Nature, position and relative size of the image formed by a convex lens for various positions of the object

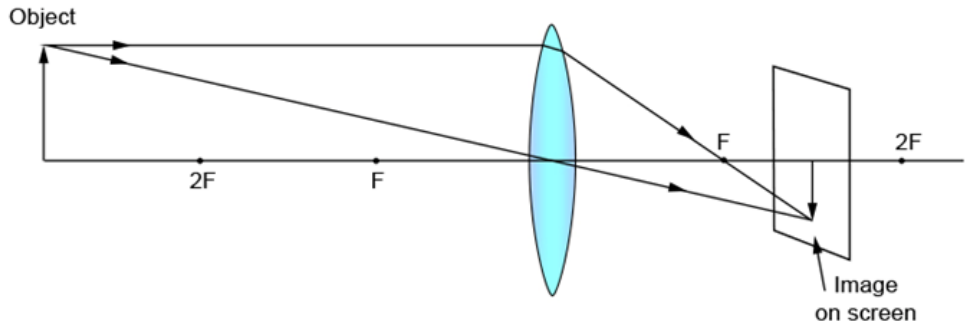
Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F_2	Highly diminished, point-sized	Real and inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real and inverted
At $2F_1$	At $2F_2$	Same size	Real and inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real and inverted
At focus F_1	At infinity	Infinitely large or highly enlarged	Real and inverted
Between focus F_1 and optical centre O	On the same side of the lens as the object	Enlarged	Virtual and erect

Object placed at infinity

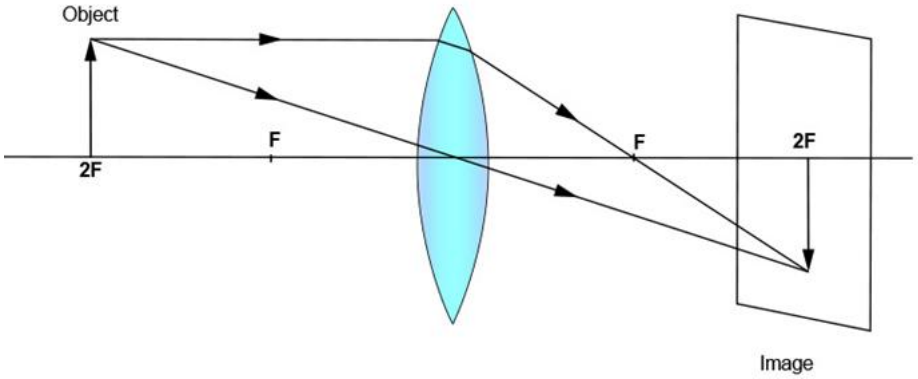


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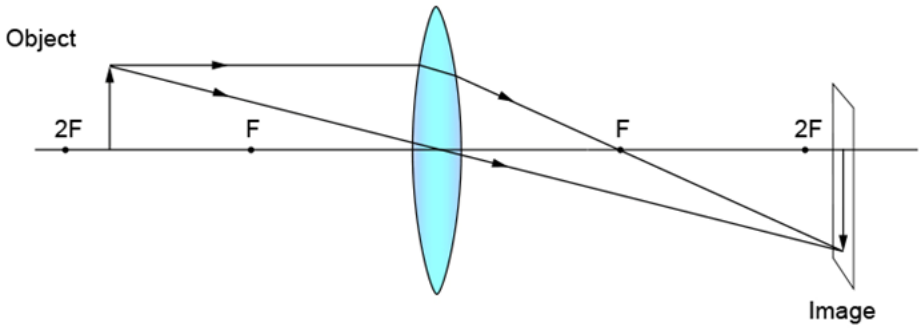
Object placed beyond $2F_1$



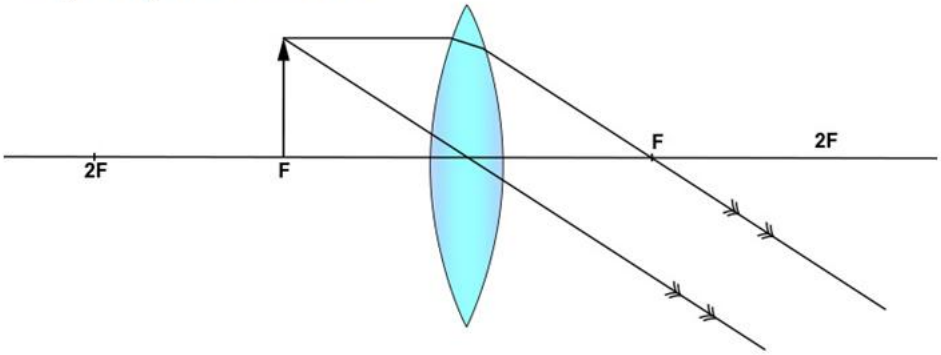
Object placed at $2F_1$



Object placed between $2F_1$ and F_1



Object placed at F_1



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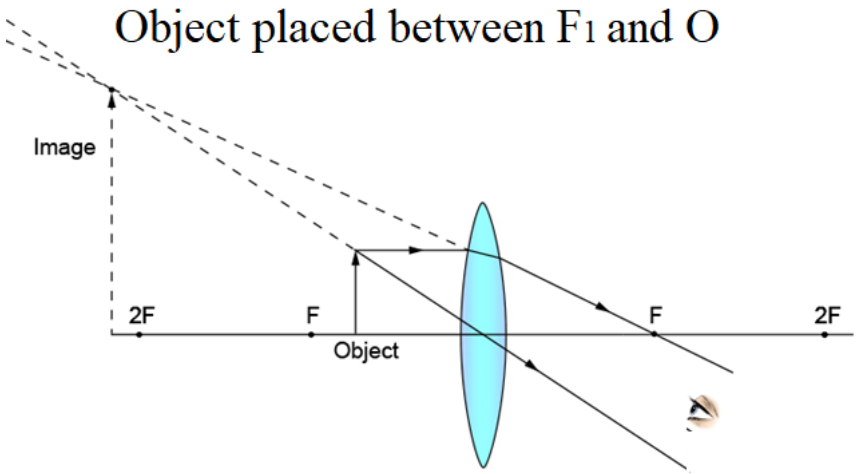
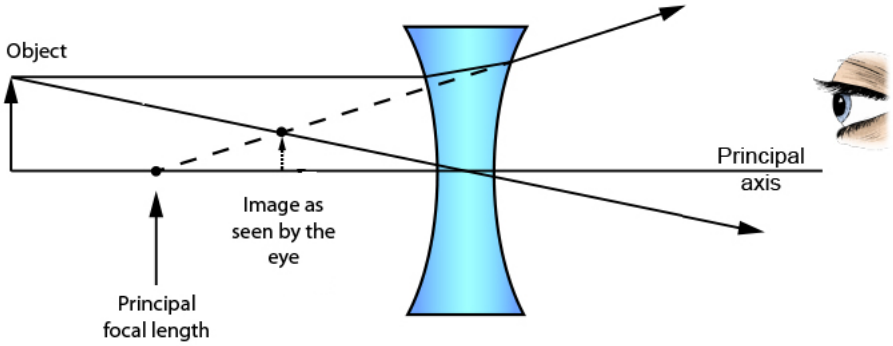


Image Formation by Concave Lens

Nature, position and relative size of the image formed by a concave lens for various positions of the object:

Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F_1	Highly diminished, point-sized	Virtual and erect
Between infinity and optical centre O of the lens	Between focus F_1 and optical centre O	Diminished	Virtual and erect



Lens formula and Magnification

Lens Formula:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Where, v = Image distance , u = Object distance , f = Focal length

Magnification :

It is the ratio of the height of image to the height of object.

$$m = \frac{Height\ of\ Image}{Height\ of\ Object} = \frac{h'}{h}$$

$$m = \frac{h'}{h} = \frac{v}{u}$$

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Key points :

- If 'm' is negative, image is real.
- If 'm' is positive, image is virtual.
- If m = 1, i.e., image is equal to object.
- If $h^i > h$ then $m > 1$ i.e., image is larger than object.
- If $h^i < h$ then $m < 1$ i.e., image is smaller than object.

Exercise :

A concave lens has focal length 30 cm. At what distance should the object be placed from the lens so that it forms an image at 20 cm from the lens ? Also, find the magnification produced by the lens.

Solution : Focal length of a lens f = -30cm , Image distance u = -20cm

Lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ OR $\frac{1}{u} = \frac{1}{v} - \frac{1}{f}$

$$\frac{1}{u} = \frac{1}{-20} - \frac{1}{(-30)}$$

$$\frac{1}{u} = \frac{1}{-20} + \frac{1}{30} = \frac{-3+2}{60} \qquad \frac{1}{u} = \frac{-1}{60}$$

$$u = -60 \text{ cm}$$

Magnification : $m = \frac{v}{u}$ $m = \frac{-20}{-60} = \frac{1}{3}$

$$m = 0.33$$

Nature of the image :

- As the m is less than 1 , image formed is smaller than the object
- As the m is +ve , image formed is virtual and erect

Power of a Lens

The degree of convergence or divergence of light rays achieved by a lens is expressed in terms of its power.

The power of a lens is defined as the reciprocal of its focal length. It is represented by the letter P.

The power P of a lens of focal length f is given by : $P = \frac{1}{f}$

The SI unit of power of a lens is **'dioptre'**. It is denoted by the letter **D**.

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1 dioptre is the power of a lens whose focal length is 1 metre.

The power of a convex lens is positive (+D) and that of a concave lens (-D) is negative.

Exercise :

Find the focal length of a lens of power -2.0 D. What type of lens is this ?

Solution : Here, P = -2.0 D

Focal of length of lens $f = \frac{1}{P} = \frac{1}{-2} = -0.5m \ or -50cm$

The type of lens is concave because the focal length is negative.

Exercise :

If the focal lengths of two lenses A and B are + 0 · 50 m and – 0 · 40 m respectively. Mention the types of these lenses in the same order.

Solution :

- + 0 · 50 m → Convex lens (because the focal length is positive)
- - 0 · 40 m → Concave lens (because the focal length is negative)

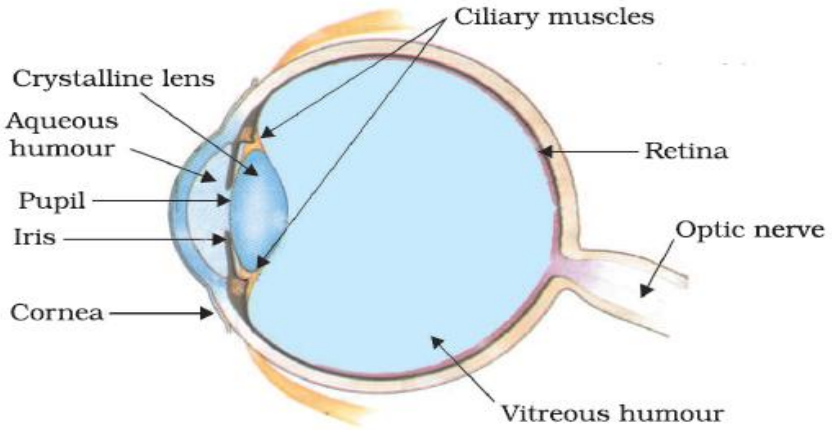
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CHAPTER-10

THE HUMAN EYE AND THE COLOURFUL WORLD

TOPIC-1 HUMAN EYE AND ITS DEFECTS

PARTS OF HUMAN EYE :



Cornea : It is the outermost, transparent part where the maximum refraction takes place.

Lens : It is composed of a fibrous, jelly like material. It provides the focused, real and inverted image of the object on the retina. The convex lens converges light at retina.

Iris : It is a dark muscular diaphragm that controls the size of the pupil.

Pupil : It is the window of the eye. It is the central aperture in iris. It regulates and controls the amount of light entering the eye.

Retina : It is a delicate membrane having enormous number of light sensitive cells.

Ciliary muscles : These muscles change the shape and size of the eye lens for focussing.

Optic Nerve : The optic nerve sends the electrical impulses from the retina, at the back of the eyes to the brain.

Far point : The maximum distance at which object can be seen clearly is far point of the eye. For a normal adult eye, its value is infinity.

Near point : It is the minimum distance at which objects can be seen most distinctively without strain. For a normal adult eye, its value is 25 cm.

Accommodation : The ability of the eye lens to adjust its focal length is called **accommodation of the eye lens.**. Focal length can be changed with the help of ciliary muscles.

- When the ciliary muscles are relaxed, the lens becomes thin. Thus, its focal length increases. This enables us to see distant objects clearly.
- When the ciliary muscles contract. This increases the curvature of the eye lens. The eye lens then becomes thicker. Consequently, the focal length of the eye lens decreases. This enables us to see nearby objects clearly.

DEFECTS OF VISION AND THEIR CORRECTION

Sometimes, the eye may gradually lose its power of accommodation. In such conditions, the person cannot see the objects distinctly and comfortably. The vision becomes blurred due to the refractive defects of the eye.

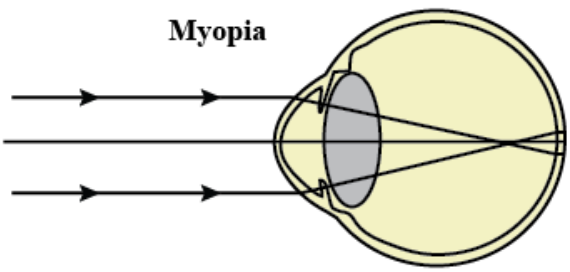
There are mainly three common refractive defects of vision. These are -

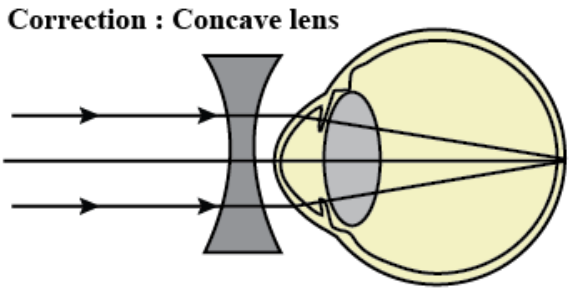
(i) myopia or near-sightedness,
(ii) Hypermetropia or farsightedness,and
(iii) Presbyopia.

These defects can be corrected by the use of suitable lenses.

Myopia

- Myopia is also known as near-sightedness. A person with myopia can see nearby objects clearly but cannot see distant objects distinctly.
- A person with this defect has the far point nearer than infinity. Such a person may see clearly upto a distance of a few meters.
- In a myopic eye, the image of a distant object is formed in front of the retina
- This defect may arise due to
  (i) excessive curvature of the eye lens, or
  (ii) elongation of the eyeball.
- This defect can be corrected by using a **concave lens** of suitable power.
- A concave lens of suitable power will bring the image back on to the retina and thus the defect is corrected.

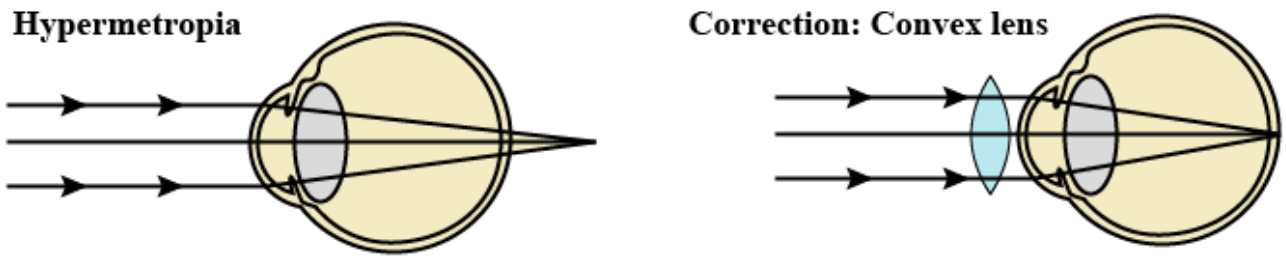




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Hypermetropia

- Hypermetropia is also known as far-sightedness. A person with hypermetropia can see distant objects clearly but cannot see nearby objects distinctly.
- The near point, for the person, is farther away from the normal near point (25 cm). Such a person has to keep a reading material much beyond 25 cm from the eye for comfortable reading.
- This is because the light rays from object are focussed at a point behind the retina
- This defect arises either because
  (i) the focal length of the eye lens is too long, or
  (ii) the eyeball has become too small.
- This defect can be corrected by using a **convex lens** of appropriate power.
- Eye-glasses with converging lenses provide the additional focusing power required for forming the image on the retina.



Presbyopia

- The power of accommodation of the eye usually decreases with ageing.
- For most people, the near point gradually recedes away. They find it difficult to see nearby objects comfortably and distinctly without corrective eye-glasses.
- This defect is called Presbyopia.
- It arises due to the gradual weakening of the ciliary muscles and diminishing flexibility of the eye lens.
- Sometimes, a person may suffer from both myopia and hypermetropia. Such people often require bi-focal lenses.
- The upper portion consists of a concave lens. It facilitates distant vision.
- The lower part is a convex lens. It facilitates near vision.

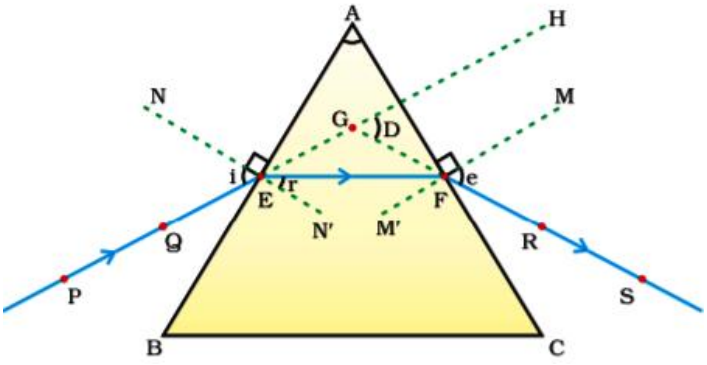
Cataract

- The crystalline lens of people at old age becomes milky and cloudy. This condition is called cataract.
- This causes partial or complete loss of vision.
- It is possible to restore vision through a cataract surgery.

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TOPIC-2 REFRACTION THROUGH PRISM AND ATMOSPHERE

REFRACTION OF LIGHT THROUGH A PRISM

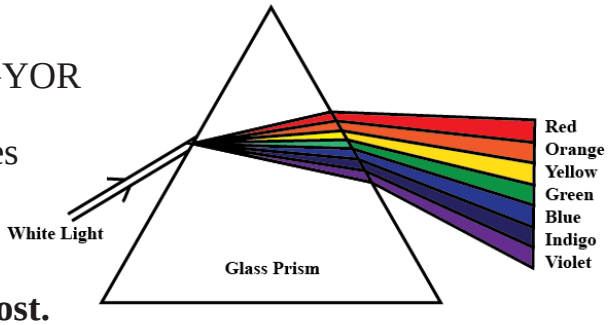


PE – Incident ray	∠i – Angle of incidence
EF – Refracted ray	∠r – Angle of refraction
FS – Emergent ray	∠e – Angle of emergence
∠A – Angle of the prism	∠D – Angle of deviation

- When a ray of light is entering from air to glass at the first surface AB. The light ray on refraction has bent towards the normal.
- At the second surface AC, the light ray has entered from glass to air. Hence it has bent away from normal.
- The peculiar shape of the prism makes the emergent ray bend at an angle to the direction of the incident ray. This angle is called the **angle of deviation**.
- In this case ∠D is the angle of deviation.

DISPERSION OF WHITE LIGHT BY A GLASS PRISM

- The splitting of light into its component colours is called **dispersion.**
- The band of the coloured components of a light beam is called its **spectrum.**
- The sequence of colours seen are Violet, Indigo, Blue, Green, Yellow, Orange and Red,
- The acronym used to remember colour - VIBGYOR



Different colours of light bend through different angles with respect to the incident ray, as they pass through a prism.

The red light bends the least while the violet the most.

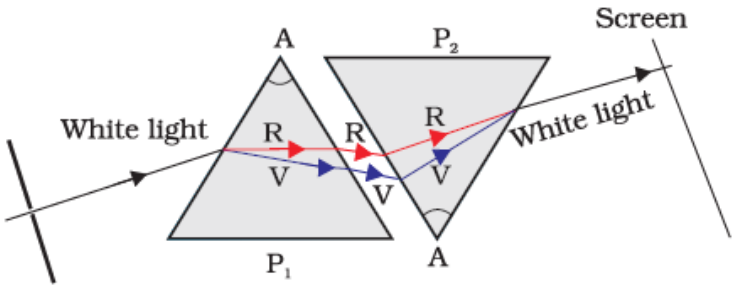
NEWTON'S EXPERIMENT

- Isaac Newton was the first to use a glass prism to obtain the spectrum of sunlight.
- He tried to split the colours of the spectrum of white light further by using another similar prism. However, he could not get any more colours.

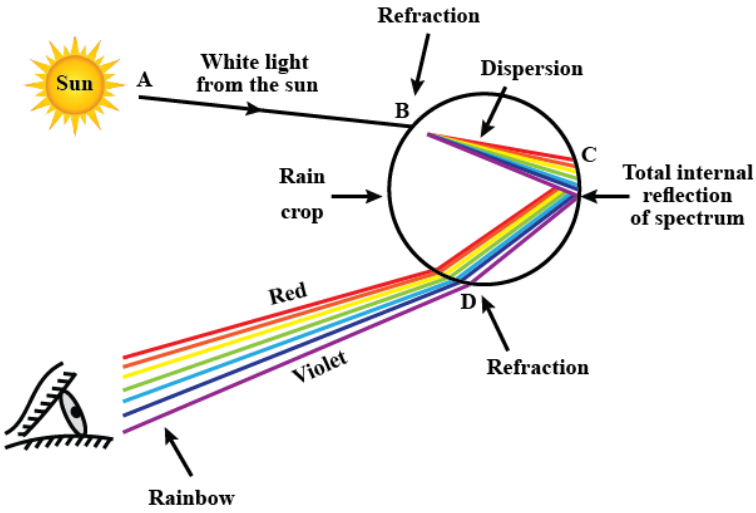
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- He then placed a second identical prism in an inverted position with respect to the first prism, This allowed all the colours of the spectrum to pass through the second prism.
- He found a beam of white light emerging from the other side of the second prism. This observation gave Newton the idea that the sunlight is made up of seven colours.

Recombination of the spectrum of white light



FORMATION OF RAINBOW



- A rainbow is a natural spectrum appearing in the sky after a rain shower.
- A rainbow is always formed in a direction opposite to that of the Sun.
- It is caused by dispersion of sunlight by tiny water droplets, present in the atmosphere.
- The water droplets act like small prisms. They refract and disperse the incident sunlight, then reflect it internally, and finally refract it again when it comes out of the raindrop.
- Due to the refraction, dispersion and internal reflection the rainbow will be formed.

ATMOSPHERIC REFRACTION

- We might have observed the apparent random wavering or flickering of objects seen above a fire.
- The air just above the fire becomes hotter than the air further up. The hotter air is lighter (less dense) than the cooler air above it, and has a refractive index slightly less than that of the cooler air.

- The refractive index of the medium is continuously changing. Due to this the apparent position of the object, as seen through the hot air, fluctuates. This wavering is thus an effect of atmospheric refraction
- We also observe such phenomenon on sunny day above the tar rod.

Twinkling of stars

- The twinkling of a star is due to atmospheric refraction of starlight.
- The starlight, on entering the earth's atmosphere, undergoes refraction continuously before it reaches the earth.

Why don't the planets twinkle?

- The planets are much closer to the earth,
- The total variation in the amount of light entering our eye is very least. Hence, there is no twinkling effect.

Differ in position of star :

- Due to the atmospheric refraction starlight bends towards normal. Hence, the apparent position of the star is slightly different from its actual position.
- The star appears slightly higher (above) than its actual position when viewed near the horizon

Advance sunrise and delayed sunset

- The Sun is visible to us about 2 minutes before the actual sunrise, and about 2 minutes after the actual sunset because of atmospheric refraction.
- By actual sunrise, we mean the actual crossing of the horizon by the Sun. Due to atmospheric refraction the apparent position of the sun will be above the horizon at sunrise and below the horizon at sunset.
- The time difference between actual sunset and the apparent sunset is about 2 minutes.
- The apparent flattening of the Sun's disc at sunrise and sunset is also due to the same phenomenon.

SCATTERING OF LIGHT

The blue colour of the sky, colour of water in deep sea, the reddening of the sun at sunrise and the sunset are some of the phenomena occurred due to scattering of light.

Tyndall Effect

- The earth's atmosphere is a heterogeneous mixture of minute particles.
- These particles include smoke, tiny water droplets, suspended particles of dust and molecules of air.

- When a beam of light strikes such fine particles, the path of the beam becomes visible.
- The phenomenon of scattering of light by the colloidal particles is called **Tyndall effect**
- The colour of the scattered light depends on the size of the scattering particles.
- Very fine particles scatter mainly **blue light** (shorter wavelength), while particles of larger size scatter **red light** (longer wavelengths).

Colour of the clear Sky

- When sunlight passes through the atmosphere, the fine particles in air scatter the blue colour (shorter wavelengths) more strongly than red. The scattered blue light enters our eyes. Hence, the sky appears blue.
- If the earth had no atmosphere, there would not have been any scattering. Then, the sky would have looked dark.
- The sky appears dark to the astronauts and the colour of the sky is black on the moon.

Colour of the Sun at Sunrise and Sunset

- Light from the Sun near the horizon passes through thicker layers of air and larger distance in the earth's atmosphere before reaching our eyes.
- Near the horizon, most of the blue light (shorter wavelengths) are scattered away by the particles. Therefore, the light that reaches our eyes is Red (longer wavelengths). This gives rise to the reddish appearance of the Sun.
- Light from the Sun overhead would travel relatively shorter distance. Hence, at noon, the Sun appears white as only a little of the blue and violet colours are scattered.
- We have observed that 'danger' signal lights are red in colour. Because, the red is least scattered by fog or smoke. Therefore, it can be seen even at a long distance.

CHAPTER-11

ELECTRICITY

TOPIC-1 ELECTRIC CURRENT , OHM'S LAW

Charge :

- Charge is a fundamental particle of matter. It may be positive and negative.
- S.I. unit of charge is **Coulomb (C).**
- 1 Coulomb Charge = Charge present on 6 x 10^{18} electrons
- Charge present on 1 electron = 1.6 x 10^{-19} C

Electric current :

The electric current is defined as the rate of flow of electric charge through any cross section of a conductor.

$$Electric\ current = \frac{Charge}{Time} \qquad OR \quad I = \frac{Q}{t}$$

SI unit of electric current = **Ampere (A)**

1 Ampere = The flow of one coulomb of charge per second

Small quantities of current are expressed in -

- Milliampere =1 mA = 10^{-3} A
- Microampere =1 μ A = 10^{-6} A

Exercise : A current of 0.5 A is drawn by a filament of an electric bulb for 10 minutes. Find the amount of electric charge that flows through the circuit.

Solution : We are given, I = 0.5 A , t = 10 min or 10 x 60 = 600 s.

Q = I x t

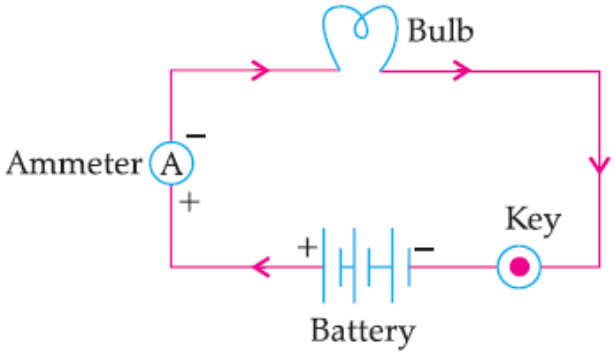
= 0.5 A x 600 s

= 300 C

Ammeter : An instrument used to measures electric current in a circuit is called Ammeter. Ammeter has low resistance and it is always connected in series in a circuit



Electric circuit : The closed path along which an electric current flows is called an 'electric circuit'.



Potential difference :

The work done to move a unit charge from one point to another is called potential difference .

It is generally known as **'Voltage'**

$$Voltage\ (V) = \frac{Work\ done (W)}{Charge\ (Q)}$$

Exercise : How much work is done in moving a charge of 2 C across two points having a potential difference 12 V?

Solution : We are given, V= 12V , Q = 2C

$V = \frac{W}{Q}$ or W = VQ

= 12 V × 2 C

= 24 J.

S.I unit of potential difference is Volt (V).

1 Volt : When 1 Joule of work is done in carrying one Coulomb charge then potential difference is called 1 Volt.

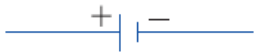
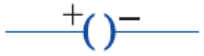
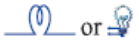
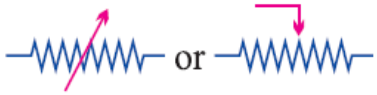
Voltmeter: It is a device to measure the potential difference. It has a high resistance and is always connected in parallel to the circuit.



- The chemical action withina cell generates the potential difference across the terminals of the cell.
- When the cell is connected to a conducting circuit element, the potential difference sets the charges in motion in the conductor and produces an electric current.

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Symbols of Some Commonly Used Components in Circuit :

Electric cell	:	
Battery	:	
Key (open)	:	
Key (closed)	:	
Wire joint	:	
Wire Crossing (without join)	:	
Electric bulb	:	
Resistance	:	
Rheostat	:	

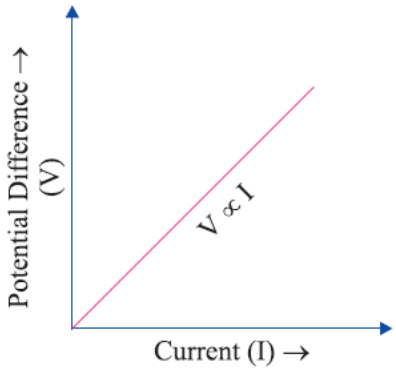
OHM'S LAW

The potential difference, V, across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it, in a constant temperature. This is called Ohm's law.

$$V \propto I$$
$$V = IR$$

R is a constant for the given metallic wire at a given temperature and is called its resistance.

V-I graph for Ohm's law : The graph between V and I is always straight line with slope equals to R.



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Resistance :

It is the property of a conductor to resist the flow of charges through it.
SI unit of resistance is **Ohm (Ω)**

If the potential difference across the two ends of a conductor is 1 V and the current through it is 1 A, then the resistance R, of the conductor is 1 Ω.

It is obvious that the current through a resistor is inversely proportional to its resistance. If the resistance is doubled the current gets halved

A component used to regulate current without changing the voltage source is called variable resistance. In an electric circuit, a device called **rheostat** is often used to change the resistance in the circuit.

Factors on which the resistance of a conductor depends :

(i) Directly proportional to the length of the conductor.
(ii) Inversely proportional to the area of cross-section.
(iii) Directly proportional to the temperature.
(iv) Depends on nature of the material.

Resistance of a uniform metallic conductor is directly proportional to its length (l) and inversely proportional to the area of cross-section (A).

$$R \ \alpha \ l \qquad and \qquad R \ \alpha \ \frac{1}{A}$$

Combining these two equations we get -

$$R \ \alpha \ \frac{l}{A} \qquad OR \qquad R = \rho \frac{l}{A}$$

where ρ (rho) is a constant of proportionality and is called the electrical resistivity of the material of the conductor.

Resistivity (ρ): The resistance offered by a wire of unit length and unit cross-sectional area is called resistivity. It is a characteristic property of the material.

The SI unit of resistivity is **Ohm-meter (Ω m).**

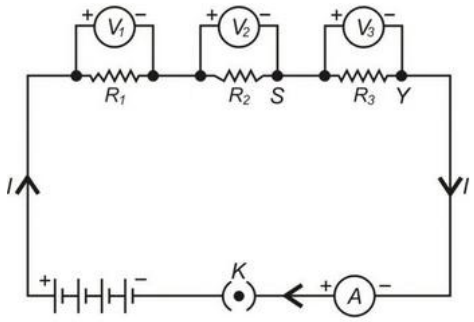
- Resistivity does not change with change in length or area of cross-section but it changes with change in temperature.
- Range of resistivity of metals and alloys is 10^{-8} to 10^{-6} Ωm.
- Range of resistivity of insulators is 10^{12} to 10^{17} Ωm.
- Resistivity of alloy is generally higher than that of its constituent metals.
- Alloys do not oxidize (burn) readily at high temperature, so they are commonly used in electrical heating devices.
- Copper and aluminium are used for electrical transmission lines as they have low resistivity.

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TOPIC-2 COMBINATION OF RESISTORS IN A CIRCUIT

Resistances in series:

When two or more resistances are connected end to end they are said to be connected in series.



- In a series combination of resistors the current is the same in every part of the circuit or the same current through each resistor
- The potential difference V is equal to the sum of potential differences V_1, V_2, and V_3.

On applying Ohm's law to the three resistors separately, we further have

$$V_1 = I R_1 \ , \ V_2 = I R_2 \ , \text{ and } V_3 = I R_3$$

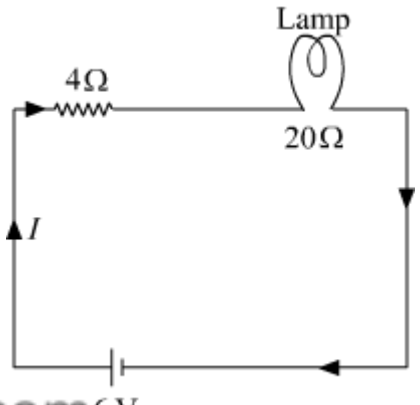
Or $I R = I R_1 + I R_2 + I R_3$

Or $Rs = R_1 + R_2 + R_3$

When several resistors are joined in series, the resistance of the combination Rs equals the sum of their individual resistances, R_1, R_2, R_3, and is thus greater than any individual resistance.

Exercise :

An electric lamp, whose resistance is 20 Ω , and a conductor of 4 Ω resistance are connected to a 6 V battery. Calculate
(a) the total resistance of the circuit, (b) the current through the circuit, and
(c) the potential difference across the electric lamp and conductor.



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Solution :

The resistance of electric lamp, $R_1 = 20\ \Omega$,
The resistance of the conductor connected in series, $R_2 = 4\ \Omega$.

(a) Then the total resistance in the circuit is

$$Rs = R_1 + R_2$$
$$Rs = 20\ \Omega + 4\ \Omega = \mathbf{24\ \Omega}$$

The total potential difference across the two terminals of the battery V = 6 V.

(b) the current through the circuit will be -

$$I = \frac{V}{R_S} = \frac{6\ V}{24\ \Omega} = \mathbf{0.25\ A}$$

(c) The potential difference across the electric lamp,

$$V_1 = 20\ \Omega \times 0.25\ A = \mathbf{5\ V}$$

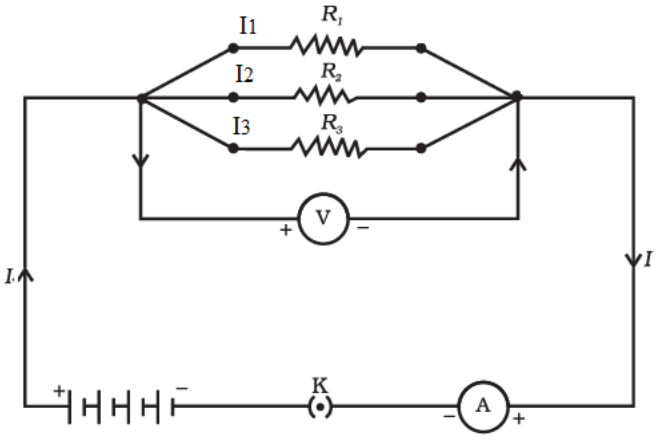
and, that across the conductor, $V_2 = 4\ \Omega \times 0.25\ A = \mathbf{1\ V}$.

Resistances in parallel :

When two or more resistances are connected across two points so that each one of them are parallel to each other, they are said to be connected in parallel.

The total current I, is equal to the sum of the separate currents through each branch of the combination.

$$I = I_1 + I_2 + I_3$$



Voltage (V) across each resistance will be same.

Let R_P be the equivalent resistance of the parallel combination of resistors.

Hence , $I = \frac{V}{R_P}$

On applying Ohm's law to each resistor, we have

$$I_1 = \frac{V}{R_1}\ , \qquad I_2 = \frac{V}{R_2}\ \ and\ \ I_3 = \frac{V}{R_3}$$

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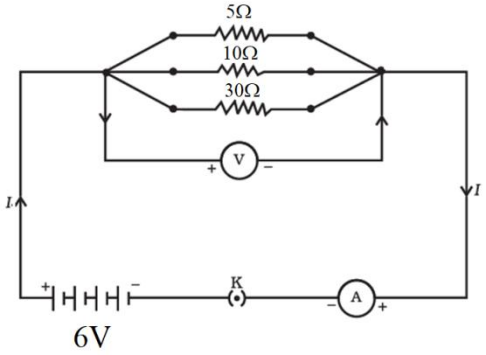
$$\frac{V}{R_P} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

The reciprocal of the equivalent resistance of a group of resistances joined in parallel is equal to the sum of the reciprocals of the individual resistances.

Exercise :

Observe the circuit diagram given below and Calculate (a) the current through each resistor, (b) the total current in the circuit, and (c) the total circuit resistance.



Solution :

$R_1 = 5\ \Omega$, $R_2 = 10\ \Omega$, and $R_3 = 30\ \Omega$.

Potential difference across the battery, V = 12 V.

The current through R_1 is $I_1 = \frac{V}{R_1} = \frac{12\ V}{5\Omega} = 2.4\ A$

The current through R_2 is $I_2 = \frac{V}{R_2} = \frac{12\ V}{10\Omega} = 1.2\ A$

The current through R_3 is $I_3 = \frac{V}{R_3} = \frac{12\ V}{30\Omega} = 0.4\ A$

The total current in the circuit, $I = I_1 + I_2 + I_3$

$= (2.4 + 1.2 + 0.4)$ A

= 4 A

The total resistance R_P is $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$= \frac{1}{5} + \frac{1}{10} + \frac{1}{30} = \frac{6+3+1}{30} = \frac{10}{30} = \frac{1}{3}$$

Thus, **Rp = 3 Ω.**

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Advantages of Parallel connection

- A parallel circuit divides the current through the electrical gadgets. This is helpful particularly when each gadget has different resistance and requires different current to operate properly.
- The total resistance in a parallel circuit is least.
- When one component fails the circuit will not breaks and other components works.

HEATING EFFECT OF ELECTRIC CURRENT

When an electric current passes through a conductor with a high resistance the conductor becomes hot after some time and produces heat. This is called **heating effect of Electric Current.**

When current **I** flowing through a resistor of resistance **R**. Let the potential difference across it be **V** and **t** be the time during which a charge **Q** flows across.
The work done in moving the charge **Q** through a potential difference **V** is **VQ**.
Therefore, the source must supply energy equal to **VQ** in time **t**.
Hence the power input to the circuit by the source is

$$P = V\frac{Q}{t} = VxI \qquad (becuase\ I = \frac{Q}{t})$$

OR For a steady current **I**, the amount of heat **H** produced in time **t** is

$$H = VIt$$

Exercise :
An electric iron consumes electric energy at a rate of 880 W when heating is at the maximum rate and 110 W when heating is at the minimum. The voltage is 220 V. What are the current and the resistance in each case.

Solution ,

We know that P = V x I , Thus, current is $I = \frac{P}{V}$

(a) When heating is at the maximum rate

$$\text{Current } I = \frac{P}{V} = \frac{880\ W}{220\ V} = 4\ A$$

$$\text{Resistance } R = \frac{V}{I} = \frac{220\ V}{4\ A} = 55\ \Omega$$

(a) When heating is at the minimum rate

$$\text{Current } I = \frac{P}{V} = \frac{100\ W}{220\ V} = 0.5A$$

$$\text{Resistance } R = \frac{V}{I} = \frac{220\ V}{0.5A} = 440\ \Omega$$

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Joule's law of heating :

The law states that - Heat produced in a resistor is,
(i) directly proportional to the square of current for a given resistance
(ii) directly proportional to resistance for a given current,
(iii) directly proportional to the time for which the current flows through the resistor

Mathematically the it is represented as $\mathbf{H = I^2Rt}$

Exercise :

The conductor having resistance 4Ω is producing 100J heat for every second. Find the current flowing through the conductor and potential difference across the conductor.

Solution : R = 4 Ω, H = 100 J, t = 1s

Formula : $H = I^2Rt$ OR $I^2 = \frac{H}{Rt}$ OR $I = \sqrt{\frac{H}{Rt}}$

$$I = \sqrt{\frac{100}{4 \times 1}} \qquad I = \sqrt{25} = 5\ A$$

Potential difference across the conductor. V = R x I

$$V = 4\ \Omega \times 5\ A = 20\ V$$

Practical Applications of Heating Effect

(a) The heating effect of electric current has many useful applications.
The electric laundry iron, electric toaster, electric oven, electric kettle and electric heater are some devices based on Joule's heating.

(b) Electric bulb :

- The electric heating is also used to produce light, as in an electric bulb.
- Tungsten is used for making bulb filaments as it is a strong metal with high melting point
- The bulbs are usually filled with chemically inactive nitrogen and argon gases to prolong the life of filament

(c) Fuse :

- Another common application of Joule's heating is the fuse used in electric circuits. It protects circuits and appliances by stopping the flow of any unduly high electric current.
- Fuse is always connected in series with live wire.
- If a current larger than the specified value flows through the circuit, the temperature of the fuse wire increases. This melts the fuse wire and breaks the circuit.

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ELECTRIC POWER

The rate at which electric energy is dissipated or consumed in an electric circuit is called **Electric power.**

The power P is given by -

$$P = V\,x\,I \quad , \quad P = \frac{I^2}{R} \quad , \quad P = \frac{V^2}{R}$$

The SI unit of electric power is **watt (W).**

Exercise :

An electric bulb is connected to a 220 V generator. The current is 0.50 A. What is the power of the bulb?

Solution : P = V x I

= 220 V x 0.50 A

= 110 W.

Exercise :

A bulb is rated 40 W; 220 V. Find the current drawn by it, when it is connected to a 220 V supply and also find its resistance.

Solution :

Current drawn by the bulb $I = \frac{P}{V} = \frac{40\ W}{220\ V} = \mathbf{0.18\ A}$

Resistance of the bulb $R = \frac{V^2}{I} = \frac{(220)^2}{0.18} = \mathbf{1210\ \Omega}$

The unit 'watt' is very small. Therefore, in actual practice we use a much larger unit called 'kilowatt'.

1 kW = 1000 watts.

Commercial unit of electric energy = kilo Watt-hour (KWh)

$1\ kWh = 3.6 \times 10^6\ J$

1 kWh = 1 unit of electric energy

Exercise :

An electric refrigerator rated 400 W operates 8 hour/day. An electric iron box rated 750 W is used for 2 hours a day. Calculate the cost of using these appliances for 30 days, if the cost of 1 kWh is Rs. 3/-.

Solution

The total energy consumed by the refrigerator in 30 days would be

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= 400 W x 8 hour/day x 30 days

= 96000 W h or = 96 kW h

The total energy consumed by the iron box in 30 days

= 750 W × 2 hour/day x 30 days

= 45000 Wh = 45 kWh

The total energy consumed by the refrigerator and iron box is

= 96 kWh + 45 kWh = 141 kWh

The sum of bill amount for 141 kWh at rate of Rs. 3 per 1 kWh is

= 141 × 3

= Rs. 423.

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CHAPTER-12

Magnetic Effects of Electric Current

TOPIC-1 ELECTRIC CURRENT , OHM'S LAW

Magnet :

Magnet is any substance that attracts iron or iron-like substances.

Properties of a magnet -

(i) Every magnet has two poles i.e., North and South.

(ii) Like poles repel each other.

(iii) Unlike poles attract each other.

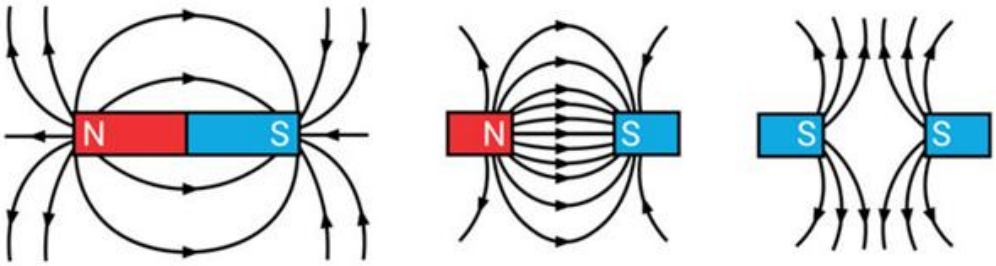
(iv) A freely suspended bar magnet aligns itself in nearly north-south direction, with its north pole towards geographical south direction.

Magnetic Field :

It is the area around a magnet in which its magnetic force can be experienced.

Magnetic Field lines :

The imaginary lines of magnetic field around a magnet are called magnetic field lines.



Properties of Magnetic Field lines :

(i) Field lines arise from North pole and end into South pole of the magnet.

(ii) Field lines are closed and continuous curves.

(iii) The density of the magnetic field lines are more in their poles.

(iv) Field lines never intersect each other as for two lines to intersect, there must be two directions of magnetic field at a point, which is not possible.

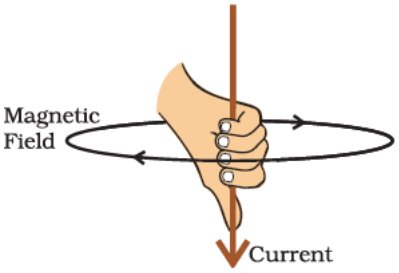
(v) Direction of field lines inside a magnet is from South to North.

(vi) The relative strength of magnetic field is shown by degree of closeness of field lines. Closer the lines, more will be the strength and farther the lines, less will be the magnetic field strength.

MAGNETIC FIELD DUE TO A CURRENT-CARRYING CONDUCTOR

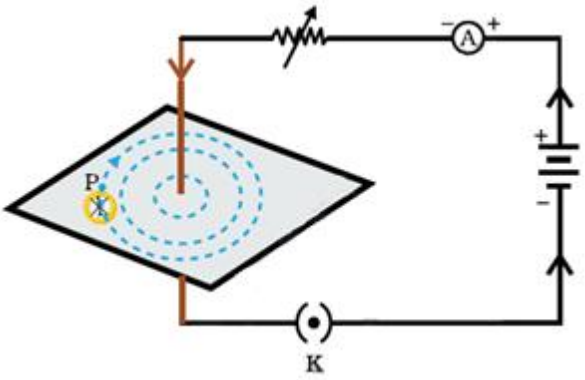
Right-Hand Thumb Rule

Right hand thumb rule states that, “If you imagine holding a current carrying wire in your right-hand with your thumb pointing towards the direction of electric current flow then the direction in which your fingers curl, gives the direction of lines of force of the magnetic field”.



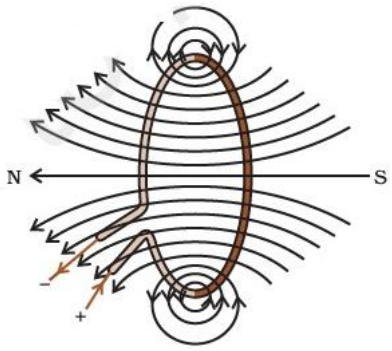
Magnetic Field due to a Current through a Straight Conductor

Tracing the pattern of magnetic field -



- Set the instruments as shown in figure .
- Place some iron fillings uniformly on the card board.
- Then close the key so that current flows through the wire.
- Gently tap the card board for few times , then we will find that the iron filings align themselves showing a pattern of concentric circles around the wire. They represent the magnetic field lines.
- The concentric circles representing the magnetic field around it would become larger and larger as we move away from the wire
- The magnitude of the magnetic field produced at a given point increases as the current through the wire increases.
- If the direction of current through the straight wire is reversed the direction of magnetic field lines also get reversed

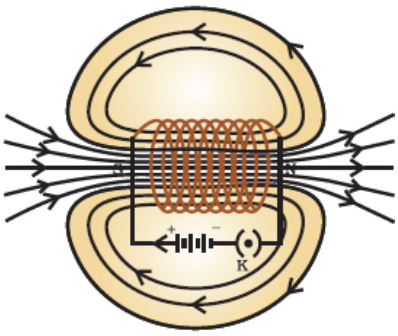
Magnetic Field due to a Current through a Circular Loop



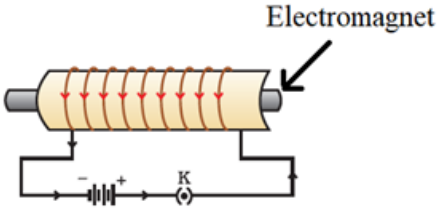
- The magnetic field pattern due to a circular coil is as shown in the given figure.
- At every point of current carrying circular loop, the concentric circles representing the magnetic field around it becomes larger and larger as we move away from the wire.
- At the center of the loop, the field appears as a straight line.
- The magnetic field produced by current-carrying circular wire depends on -
  (1) Amount of current flowing through the wire,
  (2) The number of turns of coil,

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Magnetic Field due to a Current in a Solenoid

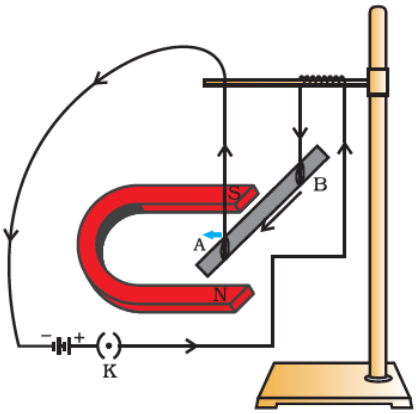


- A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder is called a solenoid.
- The pattern of the magnetic field lines around a current-carrying solenoid can be compare with the pattern of the field with the magnetic field around a bar magnet
- One end of the solenoid behaves as a magnetic north pole, while the other behaves as the south pole.
- The field lines inside the solenoid are in the form of parallel straight lines. This indicates that the magnetic field is the same at all points inside the solenoid. That is, the field is uniform inside the solenoid.
- The strength of magnetic field produced by a solenoid can be increased by
  (a) Increasing the number of turns of the coil.
  (b) Increasing the current flowing through the coil.



A strong magnetic field produced inside a solenoid can be used to magnetise a piece of magnetic material, like soft iron, when placed inside the coil. The magnet so formed is called an **electromagnet.**

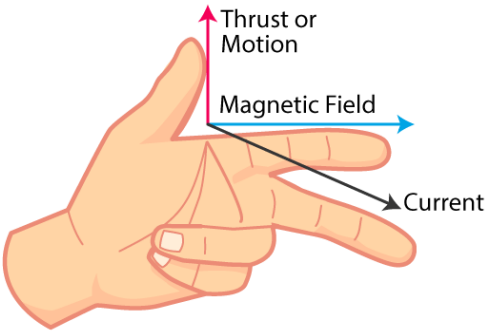
FORCE ON A CURRENT-CARRYING CONDUCTOR IN A MAGNETIC FIELD



- Force is exerted on the current-carrying aluminium rod when it is placed in a magnetic field.
- The direction of force is reversed when the direction of current through the conductor is reversed.
- If we interchange the two pole of magnet then the direction of force acting on the current-carrying rod gets reversed.
- It shows that the direction of the force on the conductor depends upon the direction of current and the direction of the magnetic field.
- The displacement of the rod is largest when the direction of current is at right angles to the direction of the magnetic field.

Fleming's Left Hand Rule :

Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular. If forefinger points in the direction of magnetic field, middle finger in the direction of current then thumb will point in the direction of motion or force.



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TOPIC-2 DOMESTIC CIRCUITS

An electric circuit consists of three main wiring components:

(i) Live wire (positive) with red insulation cover.
(ii) Neutral wire (negative) with black insulation cover.
(iii) Earth wire with green insulation cover.

Domestic circuits are of two types -

(i) A circuit with 15 A rating, suitable for appliances with higher power ratings such as geysers, air coolers, etc.
(ii) A circuit with 5 A rating for bulbs, fans, etc.

Safety devices :

Electric fuse is an important safety device used in domestic circuits. It prevents damage to the appliances and the circuit due to overloading.

Earth wire is another important safety device used in domestic circuits. It protects us from electric shock in case of leakage of current especially in metallic body appliances. It provides a low resistance path for current in case of leakage of current.

Faults in the domestic circuits :

(i) Short-circuit : when the live wire and the neutral wire come into direct contact. In such a situation, the current in the circuit abruptly increases. This is called short-circuiting.

(ii) Overloading of an electric circuit : The overheating of electrical wire in any circuit due to flow of a large current through it is called overloading of the electrical circuit.

Reasons for the overloading :

- Overloading can occur when the live wire and the neutral wire come into direct contact.
- Overloading can also occur due to an accidental hike in the supply voltage.
- Overloading is caused by connecting too many appliances to a single socket.

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